

IMPACT OF NUTRITION ON ACUTE PHYSIOLOGICAL RESPONSES TO RESISTANCE TRAINING IN HYPOXIA

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INTRODUCTION

Resistance training in normobaric hypoxia (RTH) is a growing strategy used to enhance muscular strength [1-3], muscular hypertrophy [2, 4, 5], and muscular endurance [6-8] greater than what is achievable after similar training in normoxia. The mechanisms causing the heightened efficacy of RTH are unknown, but may relate to augmented metabolic stress, characterised by greater accumulation of blood lactate and hydrogen ions [8-11]. This increased metabolic stress may stimulate anabolic hormone secretion promoting skeletal muscle growth [8-11]. However, existing investigations have only utilised fasted protocols to eliminate nutritional interference [12, 13]. This represents a significant limitation in ecological validity, as athletes and the general population rarely train in a fasted state, and reduced carbohydrate availability may impair exercise performance [10, 14]. Nutritional status fundamentally alters exercise metabolism, but no research has compared fasted and fed participants during RTH. Therefore, the primary objective of this study was to characterise the acute physiological and metabolic responses to RTH in an ecologically valid, fed state.

METHODS

Eight healthy, recreationally active males completed a randomised, single-blinded crossover intervention under three conditions: fasted normoxia (fraction of inspired oxygen; $FiO_2=0.209$), fasted hypoxia ($FiO_2=0.144$), and fed hypoxia ($FiO_2=0.144$). In the fed trial, participants consumed a standardised carbohydrate and protein-rich meal (oats, milk, and whey protein) one hour before arrival. The resistance training protocol consisted of 4x10 repetitions at 50% of estimated 1RM for four exercises (bench press, back squat, upright row, and trap-bar deadlift) with 60-second inter-set rest intervals. Oxygen saturation (SpO_2), heart rate, ratings of perceived exertion (RPE), and continuous respiratory measures (VO_2 , VCO_2 , RER) were monitored throughout the sessions. Venous blood samples were collected pre-exercise and at 0, 30, and 60 minutes post-exercise to analyse blood lactate, glucose, and insulin concentrations. Data were analysed using repeated measures mixed-effects ANOVAs with Bonferroni post-hoc pairwise comparisons.

EXPECTED OUTCOMES

This study will replicate fasted RTH protocols and determine if the physiological responses persist in a fed state. Greater postprandial physiological responses would indicate a synergistic potentiation between feeding and hypoxia on sympathetic activity and carbohydrate oxidation, supporting pre-exercise meals to further augment metabolic stress without performance decrements. Conversely, equivalent acute metabolic responses between the fed and fasted conditions would confirm fasting is not required for RTH, proving RTH remains a robust method for daily athletic training in 'real world' settings.

RESULTS, DISCUSSION AND CONCLUSION

Data is currently being analysed and will be presented on the day of the symposium.

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